Vejledende besvarelse

INSTITUT FOR MATEMATISKE FAG
AALBORG UNIVERSITET

Delopgave 1

(i) Ved at skrive $V(x_1, x_2, x_3) = \frac{q}{4\pi\varepsilon_0}(x_1^2 + x_2^+ x_3^2)^{-1/2}$ og anvende kædereglen får vi at

$$\frac{\partial}{\partial x_1} V(x_1, x_2, x_3) = \frac{q}{4\pi\varepsilon_0} \frac{-1}{2} (x_1^2 + x_2^2 + x_3^2)^{-3/2} 2x_1 = -\frac{qx_1}{4\pi\varepsilon_0 |\mathbf{x}|^3}.$$

De andre to partielle afledede findes på præcis samme måde. Den eneste forskel er at x_1 i byttes ud med x_2 og x_3, hhv. Da $\nabla V(\mathbf{x}) = \left[\frac{\partial}{\partial x_1} V(\mathbf{x}), \frac{\partial}{\partial x_2} V(\mathbf{x}), \frac{\partial}{\partial x_3} V(\mathbf{x})\right]^t$ er det derfor klart at $\mathbf{E}(\mathbf{x}) = -\nabla V(\mathbf{x})$.

(ii) Ved at bruge vinket er det klart at $V(\gamma(s))$ er en stamfunktion til $-\mathbf{E}(\gamma(s)) \cdot \gamma'(s)$. Dermed giver integralregningen (nærmere bestemt Analysens fundamentalsætning) at

$$\int_0^1 -\mathbf{E}(\gamma(s))\, ds = [V(\gamma(s))]_0^1 = V(\gamma(1)) - V(\gamma(0)) = V(\mathbf{x}) - V(\mathbf{y}).$$

Ved at omskrive overstående identitet fås

$$V(\mathbf{x}) = V(\mathbf{y}) - \int_0^1 \mathbf{E}(\gamma(s)) \cdot \frac{d\gamma}{ds}(s)\, ds. \tag{1}$$

(iii) Da $\mathbf{F}(\mathbf{x}) = \left[\frac{\partial}{\partial x_1} G(\mathbf{x}), \frac{\partial}{\partial x_2} G(\mathbf{x}), \frac{\partial}{\partial x_3} G(\mathbf{x})\right]^t$ følger det fra definitionen af $\mathrm{curl}(\mathbf{F})(\mathbf{x})$ at

$$\mathrm{curl}(\mathbf{F})(\mathbf{x}) = \left[\frac{\partial}{\partial x_2}\frac{\partial G}{\partial x_3} - \frac{\partial}{\partial x_3}\frac{\partial G}{\partial x_2},\ \frac{\partial}{\partial x_3}\frac{\partial G}{\partial x_1} - \frac{\partial}{\partial x_1}\frac{\partial G}{\partial x_3},\ \frac{\partial}{\partial x_1}\frac{\partial G}{\partial x_2} - \frac{\partial}{\partial x_2}\frac{\partial G}{\partial x_1}\right]^t.$$

Ved at bruge vinket ses at hver indgang i $\mathrm{curl}(\mathbf{F})(\mathbf{x})$ er lig 0.

(iv) Pr. definition har vi at

$$\mathrm{curl}(\mathbf{F})(\mathbf{x}) = \left[\frac{\partial F_3}{\partial x_2} - \frac{\partial F_2}{\partial x_3},\ \frac{\partial F_1}{\partial x_3} - \frac{\partial F_3}{\partial x_1},\ \frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2}\right]^t.$$

og indsætter vi dette i definitionen af divergensen fås

$$\mathrm{div}(\mathrm{curl}(\mathbf{F}))(\mathbf{x}) = \frac{\partial}{\partial x_1}\left(\frac{\partial F_3}{\partial x_2} - \frac{\partial F_2}{\partial x_3}\right) + \frac{\partial}{\partial x_2}\left(\frac{\partial F_1}{\partial x_3} - \frac{\partial F_3}{\partial x_1}\right) + \frac{\partial}{\partial x_3}\left(\frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2}\right).$$

Skriver man hele udtrykket ud, og anvender vinket fra forrige opgave fås at $\mathrm{div}(\mathrm{curl}(\mathbf{F}))(\mathbf{x}) = 0$.

(v) For at forsimple notationen definerer vi $\mathbf{A}(\mathbf{x}) = \text{curl}(\mathbf{F})(\mathbf{x})$. Dermed har vi at

$$\mathbf{A}(\mathbf{x}) = \left[\frac{\partial F_3}{\partial x_2} - \frac{\partial F_2}{\partial x_3}, \frac{\partial F_1}{\partial x_3} - \frac{\partial F_3}{\partial x_1}, \frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2}\right]^t.$$

Kombineres dette med definitionen af rotor fås at

$$\begin{aligned}
\text{curl}(\mathbf{A})(\mathbf{x}) &= \left[\frac{\partial A_3}{\partial x_2} - \frac{\partial A_2}{\partial x_3}, \frac{\partial A_1}{\partial x_3} - \frac{\partial A_3}{\partial x_1}, \frac{\partial A_2}{\partial x_1} - \frac{\partial A_1}{\partial x_2}\right]^t \\
&= \begin{bmatrix} \frac{\partial}{\partial x_2}\left(\frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2}\right) - \frac{\partial}{\partial x_3}\left(\frac{\partial F_1}{\partial x_3} - \frac{\partial F_3}{\partial x_1}\right) \\ \frac{\partial}{\partial x_3}\left(\frac{\partial F_3}{\partial x_2} - \frac{\partial F_2}{\partial x_3}\right) - \frac{\partial}{\partial x_1}\left(\frac{\partial F_2}{\partial x_1} - \frac{\partial F_1}{\partial x_2}\right) \\ \frac{\partial}{\partial x_1}\left(\frac{\partial F_1}{\partial x_3} - \frac{\partial F_3}{\partial x_1}\right) - \frac{\partial}{\partial x_2}\left(\frac{\partial F_3}{\partial x_2} - \frac{\partial F_2}{\partial x_3}\right) \end{bmatrix} \\
&= \begin{bmatrix} \frac{\partial^2 F_2}{\partial x_2 \partial x_1} - \frac{\partial^2 F_1}{\partial x_2^2} - \frac{\partial^2 F_1}{\partial x_3^2} + \frac{\partial^2 F_3}{\partial x_3 \partial x_1} \\ \frac{\partial^2 F_3}{\partial x_3 \partial x_2} - \frac{\partial^2 F_2}{\partial x_3^2} - \frac{\partial^2 F_2}{\partial x_1^2} + \frac{\partial^2 F_1}{\partial x_1 \partial x_2} \\ \frac{\partial^2 F_1}{\partial x_1 \partial x_3} - \frac{\partial^2 F_3}{\partial x_1^2} - \frac{\partial^2 F_3}{\partial x_2^2} + \frac{\partial^2 F_2}{\partial x_2 \partial x_3} \end{bmatrix}.
\end{aligned}$$

Vi husker at

$$\text{div}(\mathbf{F})(\mathbf{x}) = \frac{\partial F_1}{\partial x_1} + \frac{\partial F_2}{\partial x_2} + \frac{\partial F_3}{\partial x_3}$$

og dermed gælder at

$$\nabla(\text{div}(\mathbf{F})) = \begin{bmatrix} \frac{\partial^2 F_1}{\partial x_1^2} + \frac{\partial^2 F_2}{\partial x_1 \partial x_2} + \frac{\partial^2 F_3}{\partial x_1 \partial x_3} \\ \frac{\partial^2 F_1}{\partial x_2 \partial x_1} + \frac{\partial^2 F_2}{\partial x_2^2} + \frac{\partial^2 F_3}{\partial x_2 \partial x_3} \\ \frac{\partial^2 F_1}{\partial x_3 \partial x_1} + \frac{\partial^2 F_2}{\partial x_3 \partial x_2} + \frac{\partial^2 F_3}{\partial x_3^2} \end{bmatrix}.$$

Ved at trække $\Delta\mathbf{F}$ fra $\nabla(\text{div}(\mathbf{F}))$ får vi at

$$\nabla(\text{div}(\mathbf{F})) - \Delta\mathbf{F} = \begin{bmatrix} \frac{\partial^2 F_2}{\partial x_1 \partial x_2} + \frac{\partial^2 F_3}{\partial x_1 \partial x_3} - \frac{\partial^2 F_1}{\partial x_2^2} - \frac{\partial^2 F_1}{\partial x_3^2} \\ \frac{\partial^2 F_1}{\partial x_2 \partial x_1} + \frac{\partial^2 F_3}{\partial x_2 \partial x_3} - \frac{\partial^2 F_2}{\partial x_1^2} - \frac{\partial^2 F_2}{\partial x_3^2} \\ \frac{\partial^2 F_1}{\partial x_3 \partial x_1} + \frac{\partial^2 F_2}{\partial x_3 \partial x_2} - \frac{\partial^2 F_3}{\partial x_1^2} - \frac{\partial^2 F_3}{\partial x_2^2} \end{bmatrix}.$$

Ved at sammenligne de udregnede formler for $\nabla(\text{div}(\mathbf{F})) - \Delta\mathbf{F}$ og $\text{curl}(\text{curl}(\mathbf{F}))(\mathbf{x})$ og anvende vinket fra opgave (iii) ses at de to vektorfelter er ens.

Delopgave 2

(i) Faradays lov giver at

$$\mathrm{curl}_{\mathbf{x}}(\mathrm{curl}_{\mathbf{x}}(\mathbf{E}))(\mathbf{x},t) = \mathrm{curl}_{\mathbf{x}}\left(-\frac{\partial}{\partial t}\,\mathbf{B}(\mathbf{x},t)\right)$$

og hvis $\mathbf{B}$ er tilstrækkelig glat så kan kommuterer $\mathrm{curl}_{\mathbf{x}}$ og $\frac{\partial}{\partial t}$ (man bruger vinket fra delopgave 1(iii)). Ombyttes rækkefølgen af de to operatorer kan man anvende Ampères lov til at få

$$\mathrm{curl}_{\mathbf{x}}\left(-\frac{\partial}{\partial t}\,\mathbf{B}(\mathbf{x},t)\right) = -\frac{\partial}{\partial t}\,\mathrm{curl}_{\mathbf{x}}(\mathbf{B})(\mathbf{x},t) = -\frac{1}{c^2}\frac{\partial^2\mathbf{E}}{\partial t^2}(\mathbf{x},t).$$

Bemærk at den sidste lighed anvender at $c^2 = \frac{1}{\varepsilon_0\mu_0}$.

(ii) Fra Gauss' lov følger det at $\mathrm{div}_{\mathbf{x}}(\mathbf{E}) = 0$. Derfor giver formlen fra delopgave 1(v) og delopgave 2(i) at

$$\Delta_{\mathbf{x}}\mathbf{E}(\mathbf{x},t) = -\mathrm{curl}_{\mathbf{x}}(\mathrm{curl}_{\mathbf{x}}(\mathbf{E}))(\mathbf{x},t) = \frac{1}{c^2}\frac{\partial^2\mathbf{E}}{\partial t^2}(\mathbf{x},t).$$

(iii) Vi har følgende partielle afledede

$$\begin{aligned}
\frac{\partial}{\partial x_j}\mathbf{E}_{\mathrm{osc}}(\mathbf{x},t) &= -\mathcal{E}\sin(\omega t - \mathbf{k}\cdot\mathbf{x})(-k_j)\\
\frac{\partial^2}{\partial x_j^2}\mathbf{E}_{\mathrm{osc}}(\mathbf{x},t) &= -\mathcal{E}\cos(\omega t - \mathbf{k}\cdot\mathbf{x})k_j^2\\
\frac{\partial}{\partial t}\mathbf{E}_{\mathrm{osc}}(\mathbf{x},t) &= -\mathcal{E}\sin(\omega t - \mathbf{k}\cdot\mathbf{x})\omega\\
\frac{\partial^2}{\partial t^2}\mathbf{E}_{\mathrm{osc}}(\mathbf{x},t) &= -\mathcal{E}\cos(\omega t - \mathbf{k}\cdot\mathbf{x})\omega^2.
\end{aligned}$$

Dermed har vi at

$$\Delta\mathbf{E}_{\mathrm{osc}}(\mathbf{x},t) = -\mathcal{E}|\mathbf{k}|^2\cos(\omega t - \mathbf{k}\cdot\mathbf{x})$$

samt at

$$\frac{1}{c^2}\frac{\partial^2}{\partial t^2}\mathbf{E}_{\mathrm{osc}}(\mathbf{x},t) = -\frac{1}{c^2}\mathcal{E}\cos(\omega t - \mathbf{k}\cdot\mathbf{x})\omega^2 = -\mathcal{E}|\mathbf{k}|^2\cos(\omega t - \mathbf{k}\cdot\mathbf{x}).$$

(iv) Dette følger direkte fra de partielle aflede fra (iii) samt at $\mathcal{E}\cdot\mathbf{k} = 0$.

(v) Vi anvender i det følgende notationen $\mathcal{E} = [e_1, e_2, e_3]^t$. Ved at bruge de partielle afledede fra (iii) har vi at

$$\mathrm{curl}(\mathbf{F})(\mathbf{x}) = \begin{bmatrix} e_3\sin(\omega t - \mathbf{k}\cdot\mathbf{x})k_2 - e_2\sin(\omega t - \mathbf{k}\cdot\mathbf{x})k_3\\ e_1\sin(\omega t - \mathbf{k}\cdot\mathbf{x})k_3 - e_3\sin(\omega t - \mathbf{k}\cdot\mathbf{x})k_1\\ e_2\sin(\omega t - \mathbf{k}\cdot\mathbf{x})k_1 - e_1\sin(\omega t - \mathbf{k}\cdot\mathbf{x})k_2\end{bmatrix} = \begin{bmatrix} e_3k_2 - e_2k_3\\ e_1k_3 - e_3k_1\\ e_2k_1 - e_1k_2\end{bmatrix}\sin(\omega t - \mathbf{k}\cdot\mathbf{x}).$$

Det er nu ikke svært at se at vektoren på højresiden er $\mathbf{k}\times\mathcal{E}$.

(vi) Fra ortogonaliteten af $\mathbf{k}$, $\mathcal{E}$ og $\mathcal{B}$ følger det at Gauss' love er opfyldt. Faradays lov følger fra (v) samt

$$\frac{\partial}{\partial t}\mathbf{B}_{\text{osc}}(\mathbf{x},t) = -\mathcal{B}\omega\sin(\omega t - \mathbf{k}\cdot\mathbf{x}) = (k\times\mathcal{E})\sin(\omega t - \mathbf{k}\cdot\mathbf{x}).$$

Til sidst følger Ampères lov fra følgende betragtning

$$\text{curl}_{\mathbf{x}}(\mathbf{B}_{\text{osc}})(\mathbf{x},t) = (\mathbf{k}\times\mathcal{B})\sin(\omega t - \mathbf{k}\cdot\mathbf{x}) = -\omega\varepsilon_0\mu_0\mathcal{E}\sin(\omega t - \mathbf{k}\cdot\mathbf{x}),$$

samt de partielle afledede udregning i (iii).

Delopgave 3 (Matlab)

(i) Dette følger direkte af simple udregninger.

(ii) Bemærk først at

$$|\gamma(s)| = (c^2s^2 + s^2 + 1^2 - 2s + s^2 + s^4 - 2s^3)^{1/2} = (1 - 2s + (c^2+2)s^2 - 2s^3 + s^4)^{1/2}$$

samt at

$$\gamma(s)\cdot\gamma'(s) = c^2s - 1 + s + s - s^2 - 2s^2 + 2s^3 = -1 + (c^2+2)s - 3s^2 + 2s^3.$$

Ved at indsætte $\gamma(s)$ i $\mathbf{E}(\mathbf{x}) = \frac{1}{|\mathbf{x}|^3}\mathbf{x}$ og udregne prikproduktet fås

$$\text{E}(\gamma(s))\cdot\gamma'(s) = \frac{-1 + (c^2+2)s - 3s^2 + 2s^3}{(1 - 2s + (c^2+2)s^2 - 2s^3 + s^4)^{3/2}}.$$

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